

Adaptive Campaign Optimization Engine

Using Contextual Bandits

DS 223 — Marketing Analytics | Group Project

Spring 2026 | American University of Armenia

Milestone 1 — Product Roadmap + MoSCoW

Timeline Overview

Milestone	Dates	Weight	Focus
M1	Apr 1–10	25%	Problem definition, roles, roadmap, UI prototype
M2	Apr 10–17	25%	Database schema, simulation engine, LinUCB model
M3	Apr 19–26	25%	FastAPI backend, Streamlit frontend, Docker integration
M4	Apr 28–May 8	20%	End-to-end testing, documentation, polish
Demo	May 14	5%	Live demonstration and presentation

Milestone 1 — Foundation (Apr 1–10)

Task	Owner	Deliverable	Priority
Problem Definition document	PM	problem_definition.pdf in repo	Must
Finalize team roles	PM	Roles table in README.md	Must

Schedule call with Karen and Artur	PM	Meeting booked	Must
Product roadmap + prioritized backlog	PM	product_roadmap.pdf in repo	Must
Join GitHub Organization + repo	All	All members confirmed	Must
UI prototype / flowchart	PM / Frontend	ui_prototype.png in repo	Must
Push deliverables to repo	PM	Problem def, roadmap, UI, requirements.txt	Must
Set up GitHub Projects board	PM	Backlog / WIP / Review / Done columns	Must
Install VSCode + Project Manager ext.	All	Confirmed by all members	Must

Milestone 2 — Data + Model (Apr 10–17)

Task	Owner	Priority
PostgreSQL schema (customers, actions, rewards, model_state)	DB Dev	Must
Docker Compose: db + pgadmin containers	DB Dev	Must
Feature engineering from UCI Online Retail II (RFM + basket diversity + order size + regularity)	DS/Analyst	Must

Response simulation layer (logistic model with action-specific coefficients for 5 actions)	DS/Analyst	Must
LinUCB implementation (NumPy)	DS/Analyst	Must
Random + heuristic baseline policies	DS/Analyst	Should
ETL container: load features into DB	Backend	Should

Milestone 3 — Backend + Frontend (Apr 19–26)

Task	Owner	Priority
FastAPI endpoints: POST /decide, POST /feedback, GET /metrics	Backend	Must
SQLAlchemy models + Pydantic schemas	Backend	Must
API container in Docker Compose	Backend	Must
Streamlit dashboard: customer table + recommended action	Frontend	Must
Cumulative reward chart (policy comparison)	Frontend	Must
Simulation controls panel (n_users, alpha, cost params)	Frontend	Should
Frontend container in Docker Compose	Frontend	Must

Milestone 4 — Integration + Polish (Apr 28–May 8)

Task	Owner	Priority
End-to-end integration test (all containers, full decision loop)	All	Must
Action distribution over time chart	Frontend	Should
Documentation (MkDocs or README)	PM	Should
Regret curve visualization	DS/Analyst	Could
Code cleanup, docstrings, final requirements.txt	All	Must
Thompson Sampling as alternative algorithm	DS/Analyst	Could

Demo — May 14

Task	Owner
Live demo: run simulation, show dashboard, compare policies	All
5-min presentation: problem → solution → results	PM
Q&A preparation	All

MoSCoW Prioritization Summary

Must Have: LinUCB model, PostgreSQL, FastAPI (POST /decide, POST /feedback, GET /metrics), Streamlit dashboard (customer table, cumulative reward chart), Docker Compose, simulation engine with 5 actions, random baseline.

Should Have: Configurable alpha parameter, heuristic baseline, multiple simulation runs, action distribution chart, simulation controls panel, documentation.

Could Have: Thompson Sampling comparison, regret curve visualization, feature normalization, budget constraints on actions.

Won't Have: Deep learning, multi-step RL, real-time production deployment, full recommendation engine, distributed data infrastructure.
